at the core of sampling in molecular dynamics. Hence we are motivated to find schemes that minimize this sampling bias introduced to the long-time averages, without significant increase to the method's computational cost.

Recall that for canonical sampling, the (stationary or invariant) average of an observable $\phi(\boldsymbol{q},\boldsymbol{p})$ is given as

$$\mathrm{Av}_\beta(\phi) = \int_{\mathcal{D}} \phi(\boldsymbol{q},\boldsymbol{p})\rho_\beta(\boldsymbol{q},\boldsymbol{p})\mathrm{d}\omega, \tag{7.1}$$

where ρ_β is the canonical measure

$$\rho_\beta(\boldsymbol{q},\boldsymbol{p}) = Z_{\mathrm{can}}^{-1} \exp\left(-\beta H(\boldsymbol{q},\boldsymbol{p})\right),$$

and Z_{can} is the canonical partition function

$$Z_{\mathrm{can}}^{-1} = \int_{\mathcal{D}} \exp\left(-\beta H(\boldsymbol{q},\boldsymbol{p})\right) \mathrm{d}\omega,$$

ensuring the integral of ρ_β is unity. In this chapter, we will abbreviate the exact average of the observable ϕ to

$$\langle\phi\rangle \overset{\mathrm{def}}{=} \mathrm{Av}_\beta(\phi),$$

and write the observed average computed in experiment (using a discretization method with step size h) as $\langle\phi\rangle_h$. The error in the average of ϕ due to the perfect sampling bias can be evaluated as

$$\text{Error due to discretization } = \left|\langle\phi\rangle - \langle\phi\rangle_h\right|.$$

We assume that the observed value of $\langle\phi\rangle_h$ is corrupted as a result of our discretization, ignoring all other sources of error in our simulation. It is this sampling bias we would like to calculate (in order to correct our computed averages), or develop numerical sampling methods that naturally minimize this error.

In Fig. 7.2 we quantify the difference between the computed distribution of $q \in \mathbb{R}$ and the canonical distribution, for the one-dimensional double well potential $U(q) = (q^2-1)^2$, using Langevin dynamics with parameters $M = \gamma = \beta = 1$. The dynamics are discretized using a standard Langevin dynamics second-order algorithm taken from [62], where we sample the distribution of q along our discretized trajectory. Each step of the algorithm requires one force evaluation, so we may think of the vertical axis in Fig. 7.2 representing the total computational work.

If we were able to solve the Langevin dynamics SDE exactly, then as we evolve the system in time the distribution will converge towards the canonical distribution, however the effects of numerical discretization are apparent in the